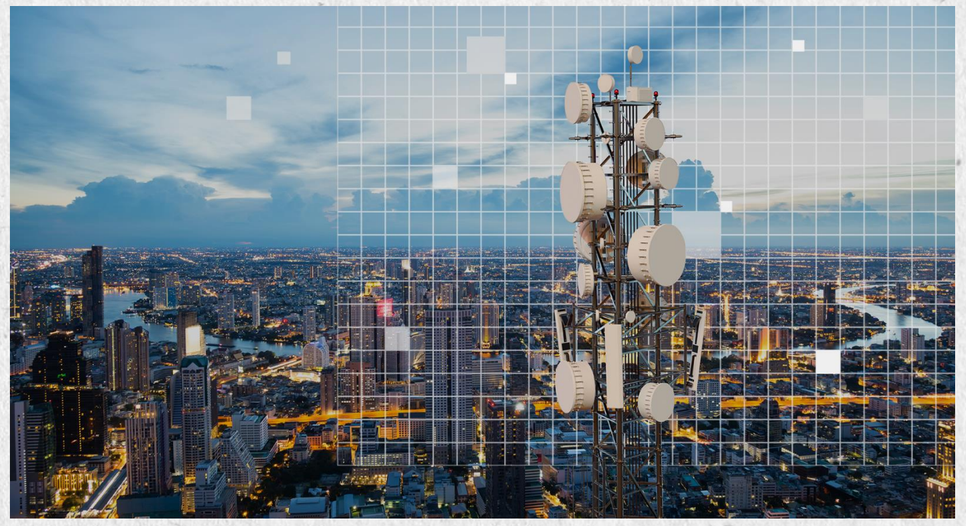
2026

From Churn Prediction to Business Growth: An AI-Powered Customer Retention Strategy

Using Machine Learning to Predict Customer Churn and Drive Revenue Growth,.



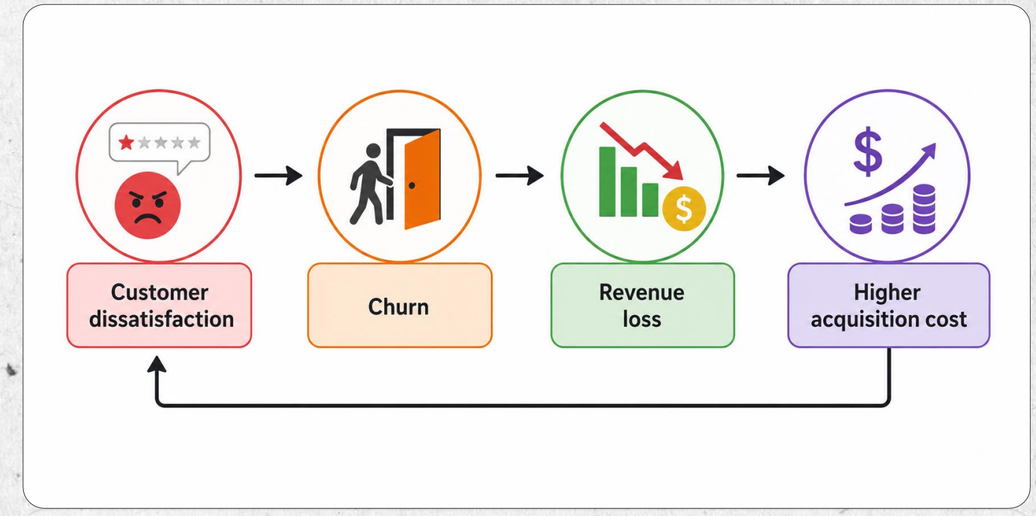
By Eman_Barakat

Problem: Customer Churn

- ✓ Telecom companies operate in a highly competitive market where customers can easily switch to another provider for better prices, device offers, or service quality.
- ✓ When customers leave, the company does not only lose monthly revenue, but also loses future customer lifetime value. At the same time, replacing lost customers requires additional marketing and acquisition costs.

Therefore, reducing churn is not only a customer service issue, it is a strategic revenue protection problem.





The business challenge

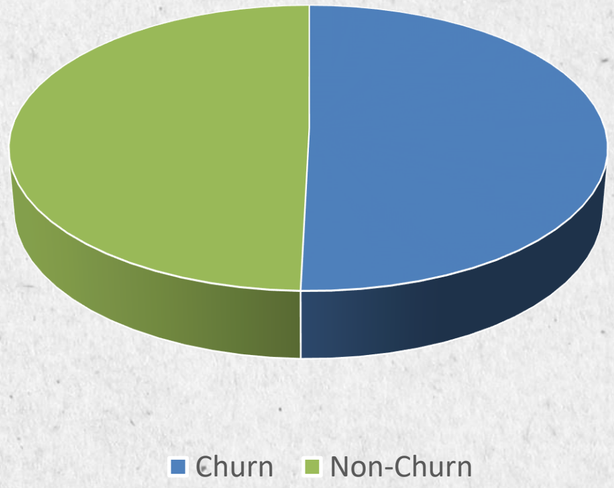
How can the company identify customers likely to leave early enough to take action?

Dataset Overview

Approximately 100 customer-related features were collected and integrated from multiple telecom data sources.

The dataset contains customer demographic, behavioral, financial, and device-related information, enabling comprehensive churn analysis.

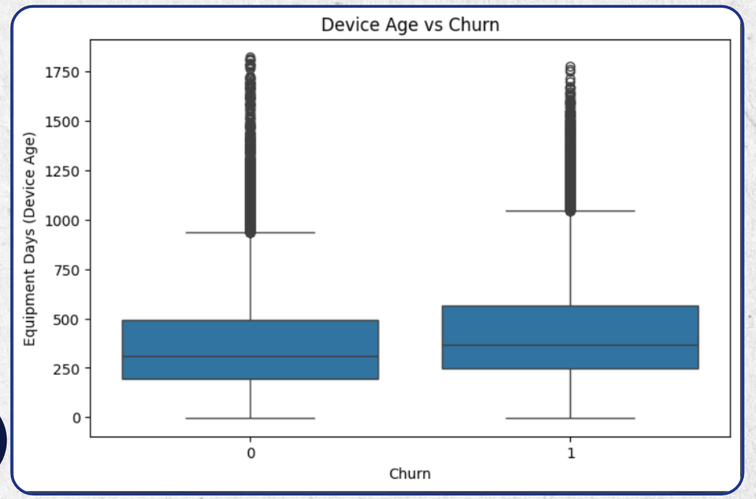
Approximately 50% of customers have churned, highlighting the importance of customer retention. Therefore, churn prediction was selected as the primary business objective to help the company proactively identify customers at risk of leaving.



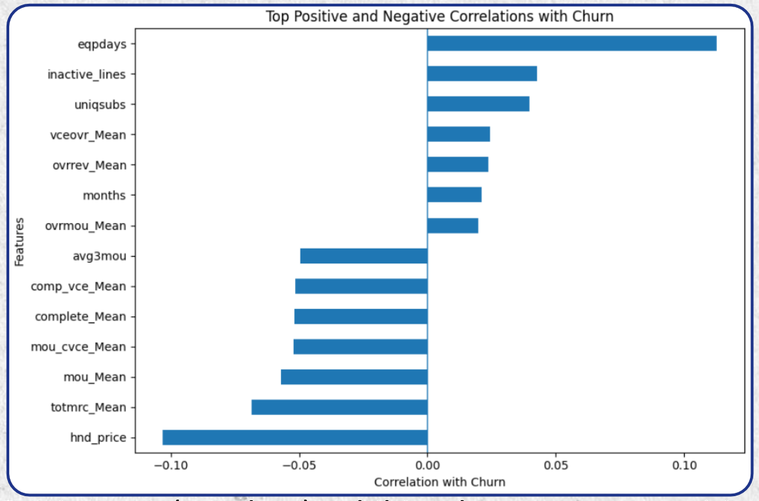
Dataset Contents

Category	Examples
Customer Profile	Income, marital status, household data
Usage Behavior	Calls, minutes of use, activity metrics
Revenue	Average revenue, total revenue
Device	Handset price, device age
Subscription	Active lines, tenure, subscriptions
Service Quality	Roaming and overage metrics

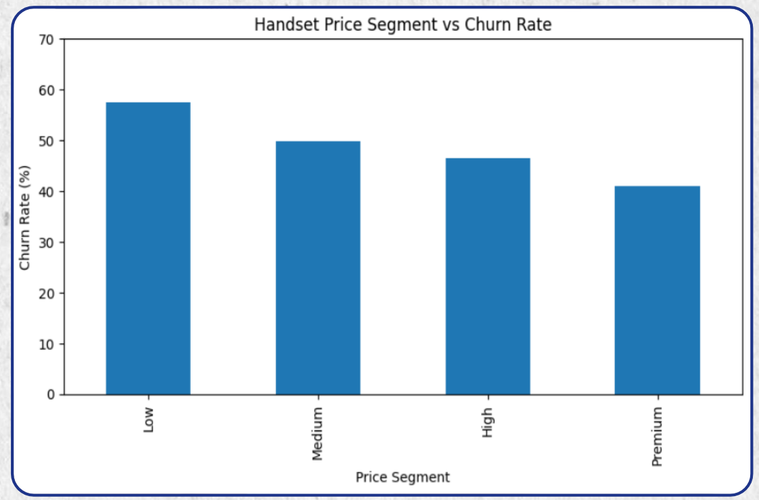
Data Visualization & Key Insights



Customers with older devices show significantly higher churn rates, suggesting device age is strongly associated with customer churn rate.



Device age (eqpdays) exhibits the strongest positive relationship with churn, because customers with higher handset prices and monthly spending tend to be less likely to leave the company.



Customers using lower-priced devices exhibit significantly higher churn rates than customers using premium devices, indicating that device value is strongly associated with customer retention.

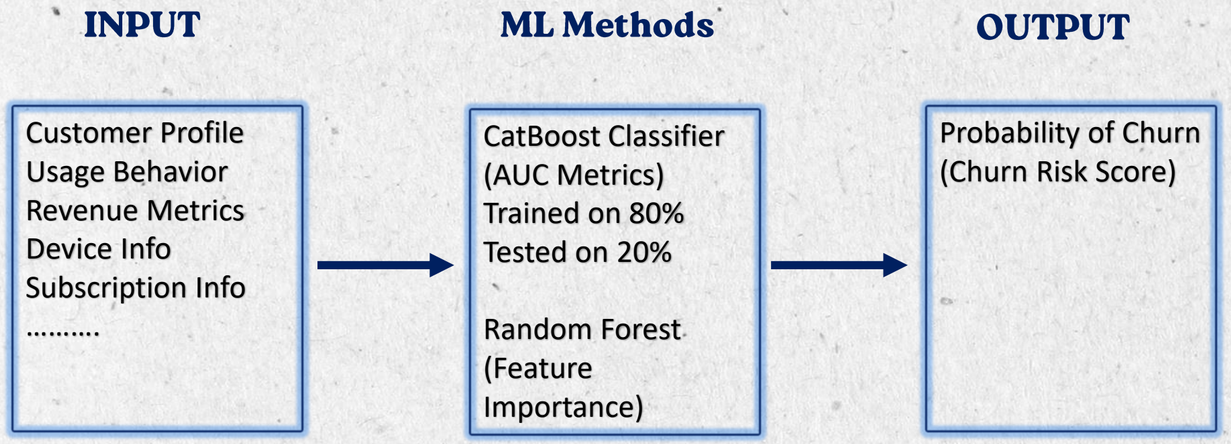
Individual features provide useful insights into customer behavior. Theses Features are insufficient to accurately identify customers who are likely to churn.



Advanced machine learning methods are necessary to uncover hidden patterns and improve churn prediction performance.

Machine Learning Model

From customer information such as usage behavior, revenue history, device information, and subscription activity, the proposed model predicts the probability of customer churn.



The CatBoost model achieved stable performance across five cross-validation folds, demonstrating its ability to identify customers at risk of churn.

To better understand the factors influencing churn, Random Forest feature importance analysis was used as an interpretability tool.

Churn Risk Score = Predicted probability of churn

Customer_ID	Actual_Churn	Predicted_Churn_Risk
1057208	1	0.9657023112733388
1011695	1	0.9567219160529968
1094819	0	0.16056857658430704
1097752	0	0.1728401912554555

The churn probability is used to rank customers from low risk to high risk and support targeted business actions.

Model Performance Evaluation

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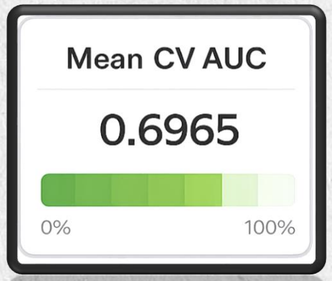
Cross Validation Results

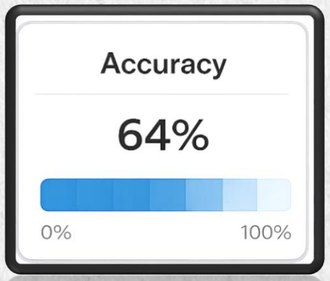
Fold	AUC
Fold 1	0.697
Fold 2	0.696
Fold 3	0.690
Fold 4	0.701
Fold 5	0.698

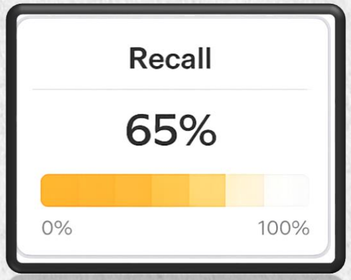
Classification Metrics

Metric	Score
Precision	0.63
Recall	0.65
F1 Score	0.64
Accuracy	0.64

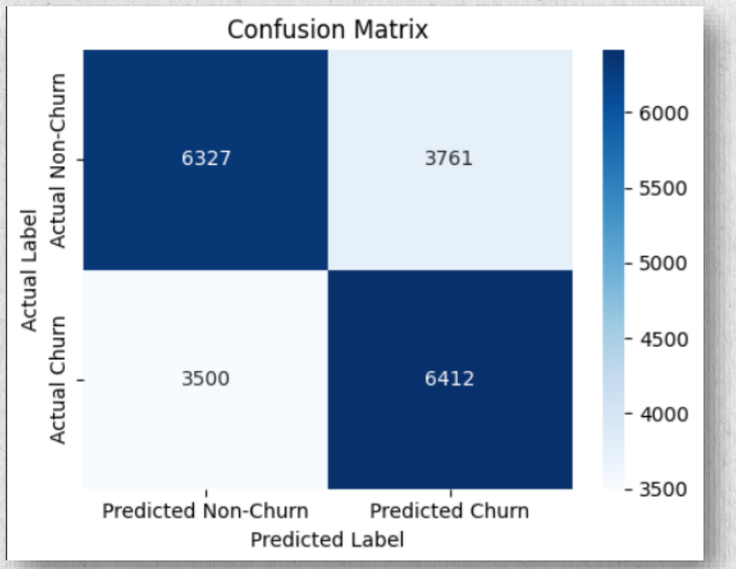
The CatBoost model achieved stable performance across all validation folds, demonstrating its ability to identify customers at risk of churn.





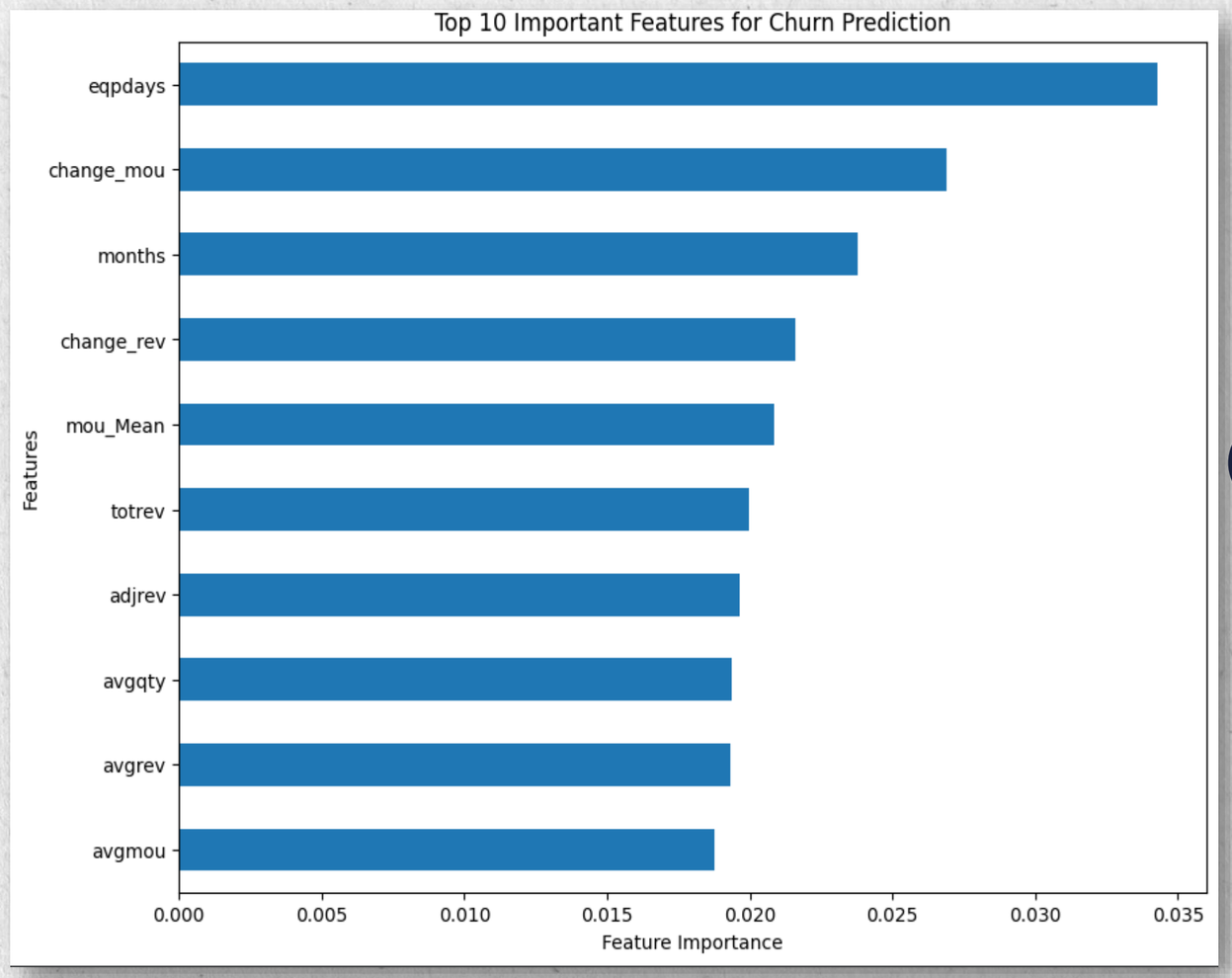


The model successfully identified 65% of churn customers while maintaining stable performance across all validation folds



Key Drivers of Churn

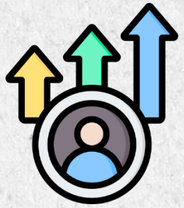
- ✓ eqpdays (Device Age) is the strongest churn indicator.
- ✓ change_mou and change_rev show that declining usage and revenue increase churn risk.
- ✓ months indicates that customer tenure influences retention behavior.
- ✓ mou_Mean highlights the importance of customer activity and engagement.



Business Proposal

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Business proposals utilizing churn prediction and customer analytics.



Device Upgrade Recommendation Engine

Enables targeted upgrade offers for customers with old or low-value devices



VIP Risk Management

Enables proactive retention actions for high-value customers with high churn risk.



Family Plan reactivation

Enables reactivation of inactive lines through personalized family bundle offers

Proposal 1:Device Upgrade Recommendation Engine

By utilizing ML, personalized device upgrade recommendations become possible

Upgrade Target
=
Old Device (Device Age > Median Device Age)
+
High Customer Value (Customer Value > Median Customer Value)

Overview

Using churn prediction and device information, customers with old devices and high churn risk can be identified automatically.

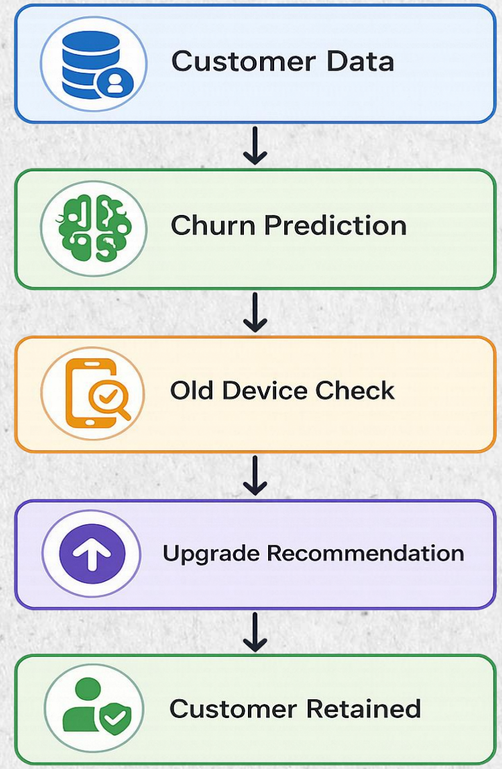
Novelty

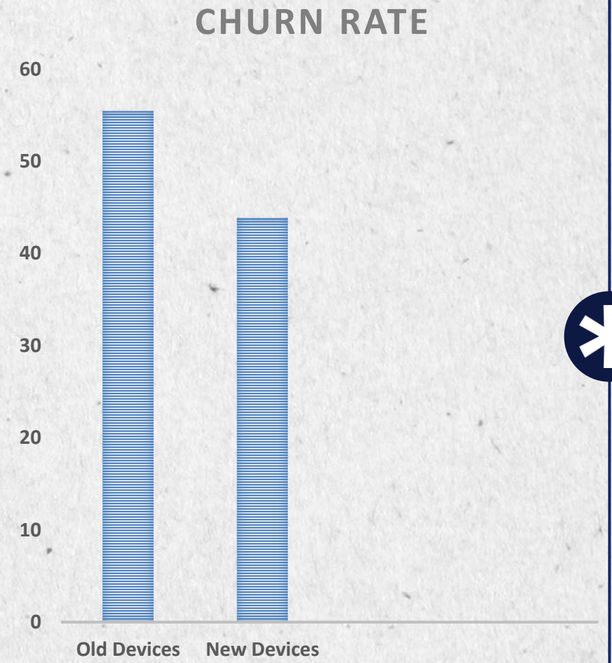
Instead of sending upgrade offers to all customers, recommendations are targeted only to customers who are most likely to leave

Effects

Customer
- Better device experience
- Improved service quality

Company
- Lower churn rate





Key Findings

Old Device Customers
Churn Rate = **55.4%**

Difference = **+11.6**

New Device Customers
Churn Rate = **43.8%**

Proposal 2: VIP Risk Management

By utilizing ML, valuable customers can be retained before revenue is lost

VIP At Risk
=
High Customer Value (Customer Value > Top 25% Highest Customer Value)
+
Old Device (Device Age > Median Device Age)

Overview

Identify customers who generate high revenue and simultaneously exhibit elevated churn risk.

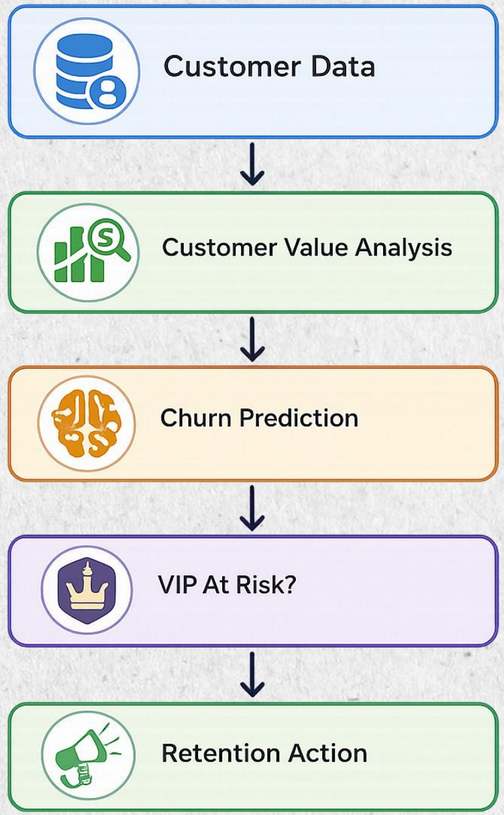
Novelty

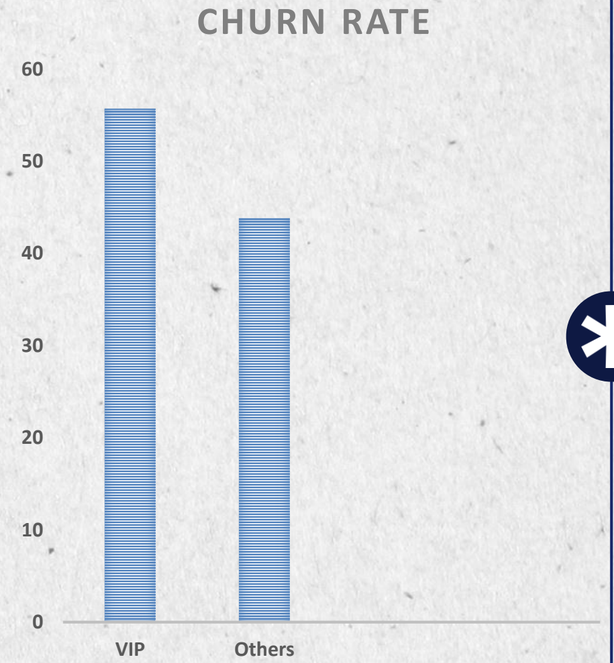
Retention efforts are focused on high-value customers rather than treating all customers equally.

Effects

Customer
- Personalized support

Company
- Revenue protection





Key Findings

VIP at RISK Customers
Churn Rate= **55.7%**

Difference = **+7 %**

Others Customers
Churn Rate= **48.7%**

Proposal 3: Family Reactivation Campaign

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By utilizing subscription analytics, inactive family lines become growth opportunities

Inactive Lines = Total Subscriptions – Active Subscriptions

Overview

Customers with inactive family lines exhibit higher churn risk and lower engagement

Novelty

Instead of focusing only on individual customers, the proposal considers household-level opportunities

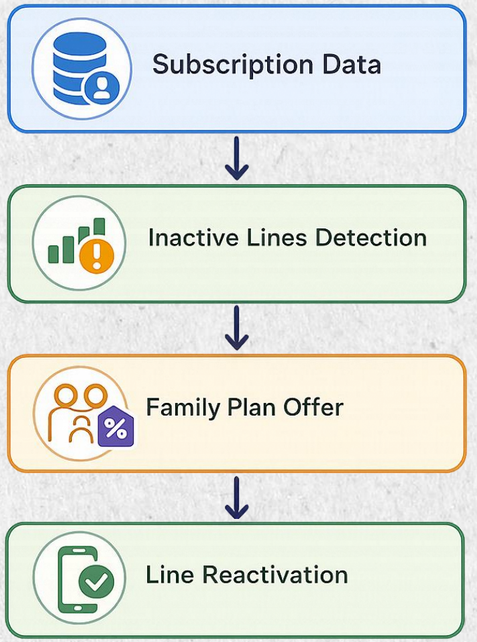
Effects

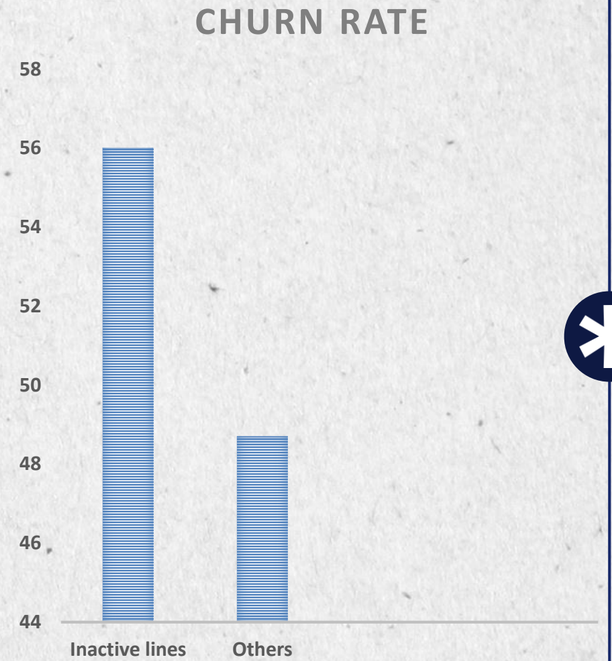
Customer

Better family plans

Company

More active subscriptions





Key Findings

Customers with Inactive Lines
Churn Rate= **56.0%**

Customers without Inactive Lines
Churn Rate= **48.5%**

Difference = **+7.5 %**

Expected Business Impact

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Customer Benefits

- ✓ Personalized offers
- ✓ Better device experience
- ✓ Improved customer satisfaction
- ✓ More suitable family plans

Business Benefits

- ✓ Reduced customer churn
- ✓ Increased customer retention
- ✓ Higher customer lifetime value
- ✓ Improved marketing efficiency
- ✓ Increased subscription revenue



Key Opportunities Identified

Opportunity	Churn Rate
Old Device Customers	55.4%
VIP At-Risk Customers	55.7%
Inactive Line Customers	56.0%

By combining churn prediction with targeted business actions, the company can reduce customer loss while increasing customer value and revenue growth

Conclusion and Future Work

Conclusion

- Customer churn is influenced by multiple behavioral, revenue, and device-related factors.
- CatBoost achieved stable predictive performance with a Mean CV AUC of 0.6965.
- Device age, revenue trends, and customer activity were identified as important indicators of churn.



Future Work

- Collect additional customer behavioral data.
- Test advanced ensemble and deep learning models.
- Develop real-time churn monitoring dashboards.
- Measure the business impact of retention campaigns after deployment.
- Continuously retrain the model using new customer data.

Rather than applying the same retention strategy to all customers, telecom companies should adopt data-driven and personalized retention programs based on churn risk and customer value

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Thank *You!